

REFLECT ON EXPERIMENT LANGUAGE

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0.1 Experiment scope

0.1.1 Which kinds of experiments do you need to run with your model to establish correctness?

To establish that the simulation behaves correctly, the project should start with simple sanity experiments that are easy to interpret. Vehicles must never take on impossible values: speeds stay at or above zero, accelerations remain within the stated physical limits, and cars do not overlap in space. In light traffic, vehicles should settle near the posted speed limit, while realistic reaction times should naturally open gaps and introduce small fluctuations in speed. Boundary conditions must also be checked. On a circular road, positions should wrap smoothly with no jumps in the trajectories. On an open road, the number of vehicles that enter and leave must be consistent with the measured flow at the downstream end. Reproducibility comes next. Running the same setup with the same random seed should give identical results, and running with different seeds should yield stable averages for key quantities such as mean speed and flow.

After these basics, the model must show the hallmark behaviors known from traffic theory and from the chosen reference systems. A brief local slowdown should trigger a phantom traffic jam that travels backward through the stream at a plausible speed close to fifteen to twenty kilometers per hour. This should appear clearly in both the time and space plot and in the density heatmap. Introducing a mix of human driving styles should increase the variability of speeds and lower the average speed, while adding autonomous vehicles should dampen fluctuations and stabilize the flow. A bottleneck zone with a reduced speed limit should produce a sustained drop in the measured outflow. Across many runs at different densities, the model should produce a classical fundamental diagram in which flow rises with density, peaks near a critical point, and then declines as congestion forms. Where comparable baselines are available, aggregated speeds, densities, and flows should be checked against them after light calibration to confirm that the results are in the right range. To further validate the implementation, we also compare our results qualitatively with the NetLogo reference model “Traffic Basic Adaptive Individuals” available at: [Link](#)

0.1.2 Which kind of experiments do you need for predictions from your model?

For prediction, the model needs a short calibration phase to align a small set of interpretable parameters with a baseline scenario. These include the distribution of human reaction times, the mix of calm, normal, and aggressive drivers, and the headway and communication delay used by autonomous vehicles. With the model calibrated, it can be run out of sample to answer practical questions. Examples include the effect of a higher share of autonomous vehicles on stability and capacity, the demand level at which the road segment reaches saturation, and the performance loss caused by longer communication delays or a more severe bottleneck. Each experiment should be repeated with multiple random seeds, and results should be reported with averages and uncertainty bands. A brief sensitivity analysis should identify which inputs drive the outcomes the most. Finally, robustness checks should vary the simulation time step and the road length to ensure that conclusions do not depend on numerical settings or on a particular geometry.

0.2 Sample experiments

In this project we use a computer simulation of cars driving on a road. Some cars are driven by human drivers with different driving styles, while others can be autonomous vehicles that react in a more controlled way. The goal of the experiments is to see whether the simulation behaves like real traffic and to use it to explore what happens under different conditions.

The first sample experiment is a baseline study on a circular road with only human drivers. We change how many cars are on the road, starting from light traffic and moving up to very dense traffic, while keeping the typical reaction times and driving styles fixed. For each traffic level we run the simulation many times with different random seeds. From each run we record the average speed, the flow of cars, the density of cars on the road and whether so-called phantom traffic jams appear. A phantom jam is a wave of stopping and starting that moves backward through the line of cars even though there is no obvious reason such as an accident. We check that the model produces a realistic fundamental diagram, where flow first increases with density, then reaches a maximum, and finally drops as congestion forms. We also check that the backward-moving jam front travels at a plausible speed between fifteen and twenty kilometers per hour and that these patterns match what is known from traffic theory and from the NetLogo reference model Traffic Basic Adaptive Individuals, which we use as our referent system.

The second sample experiment looks at the effect of autonomous vehicles on traffic stability near the critical density where the flow is close to its maximum. We choose a traffic level close to the peak of the fundamental diagram and vary how many of the cars are autonomous, from none at all up to a high share of autonomous vehicles. The total demand and the shape of the road stay the same. For each share of autonomous vehicles we also try different choices of following distance and different delays in communication between the autonomous vehicles. Again we repeat each setting many times with different random seeds. For every case we measure how much the speed varies over time, how often stop and go waves appear and how long they last, the average travel time for a car to pass through the road segment, and the resulting flow. This set of experiments shows how a higher share of autonomous vehicles and quicker communication can smooth out fluctuations, reduce phantom jams and in some cases increase the overall capacity of the road.

The third sample experiment focuses on an open road with a bottleneck zone, for example a short part of the road where the speed limit is lower. Here we fix a realistic mix of calm, normal and aggressive human drivers, together with a chosen fraction of autonomous vehicles, and then change two things. First, we change how many cars per hour try to enter the road from upstream. Second, we change how strong the bottleneck is by using different reduced speed limits in the bottleneck zone. For each combination of upstream demand and bottleneck strength we run the simulation many times with different random seeds. From these runs we collect time series of how many cars enter and leave the road, how long the resulting queue becomes, and how the density of cars is distributed along the road. These results tell us at which demand level the system becomes saturated, how large the capacity drop is because of the bottleneck, and how the presence of autonomous vehicles influences the build-up and clearing of queues.

The individual results are combined in two main steps. First, within each group of experiments, we average over the different random seeds for every chosen parameter setting. This gives smooth estimates of key quantities such as mean speed, flow, travel time and stability indicators, and also allows us to add uncertainty bands that show the variation between runs. Second, we compare the aggregated outcomes across the different experiment groups. The baseline circular-road experiments define and validate the fundamental diagram and the

typical phantom jam behavior. The experiments with mixed human and autonomous traffic then show how stability and capacity change when the share of autonomous vehicles and their reaction settings are varied. Finally, the bottleneck experiments extend these findings to a more realistic open-road situation with an entrance, an exit and a local restriction. Taken together, these three sets of experiments provide a coherent picture of how the model behaves under different conditions and how robust the conclusions are to randomness in the simulation and to reasonable changes in the input parameters.

0.3 Experiments background

This project uses four simple traffic stories to guide the experiments: a calm road with no disturbances, a short disturbance that creates a phantom jam, a crowd of drivers with different temperaments, and a short work zone that slows everyone down. In the report these stories were shown as snapshots such as a time and space plot of vehicle trajectories, a heatmap of how crowded the road becomes over time and distance, and the familiar curve that links flow and density. The experiments take each story and turn it into a precise setup that the simulator can run and measure.

The calm road story becomes a circular road with a chosen number of cars and no external events. This lets the team check basic behavior and build the fundamental diagram. The circular road is a clean way to study density because the number of cars on the loop directly fixes how crowded it is. The snapshots here are the smooth lines in the time and space plot and a rising then falling flow curve as density increases. In the experiment this story is generalized by sweeping the number of cars from light to heavy traffic and by repeating each setting with several random seeds so the averages are stable.

The phantom jam story starts from the same circular road but adds one short disturbance after the warmup. In the report this was shown as a wave that travels backward through the traffic. The experiment turns that idea into an exact change in speed at a fixed location and time, and then measures how the wave moves using the time and space plot and the density heatmap. The team checks that the wave speed falls in the expected range. This story is also generalized by trying different traffic densities so that the sensitivity to crowding is visible.

The driver variation story becomes a population of human drivers with different reaction times and different following styles, sometimes joined by autonomous vehicles with quicker and more consistent responses. In the snapshots this shows up as larger or smaller spread in the speeds inside the observation zone. The experiment makes this concrete by drawing the human parameters from simple distributions and by setting the share of autonomous vehicles to several levels. The outcomes are compared across levels to see how diversity and automation change stability, mean speed, and flow.

The bottleneck story moves from a circular road to an open road with an entry and an exit and a short zone where the speed limit is lower. In the report this was shown as a queue that forms upstream of the restriction and drains when demand falls. The experiment fixes a realistic mix of human and autonomous vehicles, then varies two dials: how many cars per hour arrive from upstream and how severe the speed reduction is inside the bottleneck. The measured snapshots are the time series of inflow and outflow, the length of the queue, and the spatial distribution of density. This setup generalizes the earlier stories because it tests the same mechanisms under more realistic boundary conditions.

In short, every scenario described in the report has a direct and testable form in the experiments. Each narrative snapshot is reproduced by a specific plot and a specific set of

measurements. The experiments also widen the scope by sweeping key inputs and repeating runs with different seeds, so that the findings do not hinge on a single picture or a single lucky run.

0.4 Experiment concepts

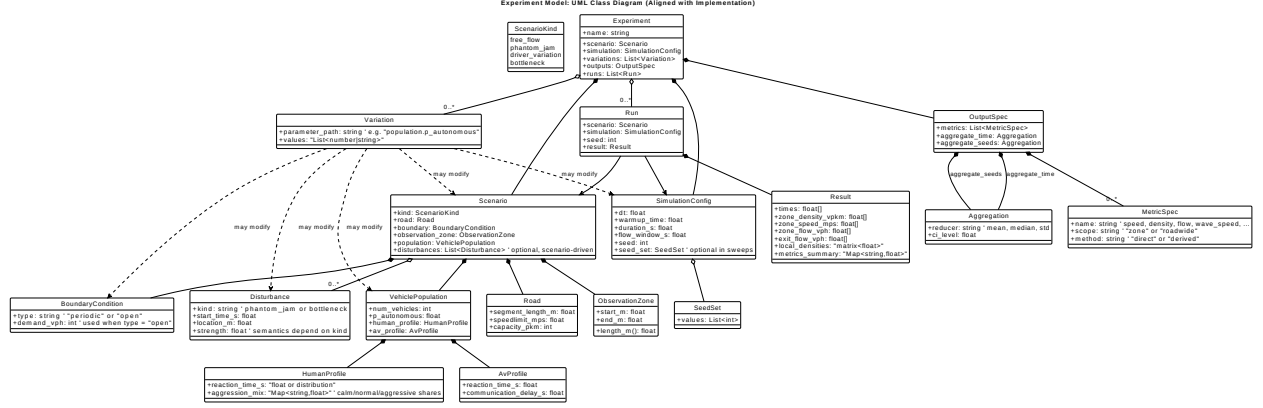


Figure 1: UML class diagram

An experiment is described in two places. One place describes the traffic world on the road, and the other place describes how the simulator is executed. The traffic world is the Scenario. It states the kind of scenario to run, chosen from free flow, phantom jam, driver variation, or bottleneck. It specifies the Road through the total segment length in meters, the speed limit in meters per second, and the nominal capacity per kilometre. It defines the BoundaryCondition by stating whether the road is periodic or open, and for open roads it also sets the upstream demand in vehicles per hour. It places the ObservationZone by giving the start and end positions in meters and therefore fixes the measured section. It defines the VehiclePopulation by giving the number of vehicles, the fraction that are autonomous, and the profiles for humans and autonomous vehicles. The HumanProfile may include a single reaction time in seconds or a distribution of reaction times, together with the mix of calm, normal, and aggressive behaviour. The AvProfile sets the reaction time in seconds and the communication delay in seconds. A scenario may also list optional Disturbance entries. A disturbance has a kind such as phantom jam or bottleneck, together with start time, location on the road, and a strength parameter whose meaning depends on the kind.

The simulator setup is the SimulationConfig. It fixes the integration time step, the warmup time, the duration in seconds, and the flow measurement window in seconds for the sliding exit-flow counter. It also fixes reproducibility. This can be done with a single seed or with a SeedSet that contains a list of integer seeds for averaging across runs. Taken together, the scenario and the simulation configuration define the conditions for one Run.

Most studies vary an input across several values. That is the role of Variation. A variation names a parameter path, such as `population.p_autonomous` or `boundary.demand_vph`, and a list of values to try. Variations expand into a grid of runs, and when a seed set is present this grid is evaluated for each seed. Each run produces a Result. A result contains the time vector, the time series measured in the observation zone, and the spatial density matrix along the road. From these series the code produces a metrics summary with the aggregated indicators used in the report. Typical metrics are average speed, density, flow and exit flow, the travel time index, wave speed, and stability indicators derived from speed variance.

The way in which results are combined is declared in the `OutputSpec`. It lists which metric specifications to report and how to aggregate them. The time aggregation rule describes the horizontal aggregation within each run, for example a time mean after warmup. The seed aggregation rule describes the vertical aggregation across seeds, for example a mean with a given confidence level. These pieces are arranged by the top-level `Experiment` object. It holds the `Scenario`, the `SimulationConfig`, the list of `Variation` items, and the `OutputSpec`, which keeps figures comparable across experiments.

0.5 Experiment syntax

Extended Backus-Naur Form (EBNF), is a formal way to write down the rules of a language, similar to how you might describe grammar rules in English class, but more precise. In EBNF, a rule may be written like this "`<thing> ::= pattern`". This means "a thing is defined as this pattern." For example, if we wanted to say that a person's name consists of a first name followed by a last name, we can write:

```
<person_name> ::= <first_name> <last_name>
```

EBNF has some special symbols:

- Square brackets [...] mean something is optional (it can appear zero or one times)
- Curly braces {...} mean something can repeat zero or more times
- A vertical bar | means "or" - you can choose one option or the other
- Parentheses (...) group things together

```
<experiment> ::= "experiment" <exp_name> "{"
                <scenario_config> <simulation_config>
                [<variation_spec>] <output_spec> "}"
```

```
<exp_name> ::= <identifier>
```

```
<scenario_config> ::= "scenario" "{"
                    "road_length" ":" <number> ";"
                    "speed_limit" ":" <number> ";"
                    "p_autonomous" ":" <number> ";" "}"
```

```
<simulation_config> ::= "simulation" "{"
                      "duration" ":" <number> ";"
                      "time_step" ":" <number> ";"
                      "warmup" ":" <number> ";"
                      "seed" ":" <seed_spec> ";" "}"
```

```
<seed_spec> ::= <number> | "range" "(" <number> "," <number> "," <number> ")"
```

The `scenario` block defines physical parameters (road geometry, speed limits, vehicle mix). The `simulation` block controls temporal aspects (duration, time step, warmup period, random seeds).

0.6 Experiment meaning

0.6.1 Formal Semantics

An experiment specification $E = (S, C, V, O)$ generates a set of simulation runs with aggregated outputs, where:

- $S = (\ell, v_{max}, p_{auto})$ defines scenario parameters: road length (m), speed limit (m/s), and autonomous vehicle fraction $\in [0, 1]$.
- $C = (T, \Delta t, T_{warmup}, \Sigma)$ defines simulation configuration: duration (s), time step (s), warmup period (s), and random seed set.
- $V = \{(p_1, V_1), \dots, (p_n, V_n)\}$ defines parameter variations, where each p_i is a parameter path and V_i is a set of values.
- $O = (M, A_h, A_v)$ defines outputs: metric set M , horizontal aggregation function A_h , and vertical aggregation function A_v .

The total number of simulation runs is:

$$N_{runs} = |\Sigma| \times \prod_{i=1}^n |V_i| \quad (1)$$

where the product term gives the number of distinct parameter configurations. Each configuration is evaluated with every seed in Σ , and results are aggregated first temporally (horizontal) then statistically (vertical).

0.6.2 Concrete Examples

Example 1: Baseline Validation

```
experiment baseline_human {
  scenario {
    road_length: 5000; speed_limit: 33.3; p_autonomous: 0.0;
  }
  simulation {
    duration: 3600; time_step: 0.1; warmup: 600; seed: 42;
  }
}
```

This experiment generates $N_{runs} = 1$ simulation: a 5 km road segment with 100% human drivers operating for one hour after a 10-minute warmup. The three metrics (throughput, travel time, density) are time-averaged within the run. This establishes baseline performance for pure human traffic and validates that the model produces realistic fundamental diagram relationships.

Example 2: Parametric Penetration Study

```
experiment autonomous_penetration_study {
  scenario {
    road_length: 5000; speed_limit: 33.3; p_autonomous: 0.0;
  }
  simulation {
    duration: 3600; time_step: 0.1; warmup: 600;
    seed: range(1, 5, 1);
  }
}
```

```
}  
variations {  
  p_autonomous: [0.0, 0.25, 0.5, 0.75, 1.0];  
}  
}
```

This experiment systematically varies autonomous vehicle penetration across five levels with five random seeds per level, generating $N_{runs} = 5 \times 5 = 25$ simulations.